Fiscal Year:	FY 2023	**Task Last Updated:**	FY 03/29/2023
PI Name:	Strock, Nicole Ph.D.		
Project Title:	Temporal Changes in Astronauts' Muscle and Cardiorespiratory Physiology Pre, During, and Post Spaceflight		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Aerobic**:Risk of Reduced Physical Performance Capabilities Due to Reduced Aerobic Capacity (2) **Muscle**:Risk of Impaired Performance Due to Reduced Muscle Size, Strength and Endurance		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	nicole.c.strock@nasa.gov	**Fax:**	FY
PI Organization Type:	GOVERNMENT	**Phone:**	
Organization Name:	KBR		
PI Address 1:	JSC B21 / 1216F		
PI Address 2:			
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	36
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2017-2018 HERO 80JSC017N0001-BPBA Topics in Biological, Physiological, and Behavioral Adaptations to Spaceflight. Appendix C
Start Date:	06/07/2021	**End Date:**	09/30/2033
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
Contact Email:	becky.brocato@nasa.gov		
Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:	New Principal Investigator (PI) as of March 2023 is Dr. Nicole Strock. Previous PIs were Dr. Eric Rivas -- and prior to Dr. Rivas, the PI was Dr. Meghan Downs. Collaborators are Scott Smith, Ph.D. and Jason Norcross, M.S.		
COI Name (Institution):	Prejean, Brian (KBR/NASA Johnson Space Center) Norcross, Jason (KBR/NASA Johnson Space Center) Schlotman, Taylor (KBR/NASA Johnson Space Center) Marshall-Goebel, Karina (NASA Johnson Space Center) Bishop, Michael (KBR/NASA Johnson Space Center)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE: Continuation of "Temporal Changes in Astronauts' Muscle and Cardiorespiratory Physiology Pre, During, and Post Spaceflight" with new Principal Investigator, Dr. Nicole Strock. Quantification of astronauts' changes in cardiorespiratory fitness and muscle performance in parallel with monitoring of pre and in-flight lifestyle habits (i.e., exercise training, nutritional intake, and sleep patterns) is needed to develop countermeasures and technologies for monitoring and mitigating crew health and performance risks during exploration class missions. The research proposed herein will profile temporal changes in astronauts' cardiorespiratory fitness and muscle mass, strength, and endurance over the course of spaceflight missions ranging from 2 months, 6 months, and up to 1 year in duration. Additionally, a statistical based extrapolation will provide predictions for changes associated with exploration missions 2-3 years in duration. To accomplish these objectives astronauts will be asked to participate in pre, in, and postflight measurement of muscular and cardiorespiratory fitness. Additionally, data collected by other "Complement of Integrated Protocols for Human Exploration Research" (CIPHER) investigators and Medical Operations testing (i.e., function testing, bone, dietary, exercise, sleep, and cognition outcomes) will be obtained via data sharing. Muscular fitness will be tested pre, in, and postflight using isometric (mid-thigh pull, leg press) and isokinetic (knee, ankle, trunk extension/flexion) methods. Cardiorespiratory fitness and related parameters will be tested pre, in, and postflight using aerobic capacity (VO2peak) testing. This proposal specifically addresses the temporal effects of spaceflight on changes in cardiorespiratory fitness and muscle function, both critical parameters in maintaining the ability to perform mission critical tasks and enabling safe human space exploration beyond low-Earth orbit. Integration of data collected during the pre, in, and postflight periods will lead to a better understanding of how to optimize exercise and non-exercise countermeasures to maintain crew health, safety, and performance during the exploration era.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	It is well established that exercise has many health benefits. This research will improve our understanding about the variability fitness improvements and how they change over time. Understanding this will improve exercise prescriptions for athletic and clinical populations on Earth.
Task Progress:	New PI for FY2023. NOTE this is continuation of "Temporal Changes in Astronauts' Muscle and Cardiorespiratory Physiology Pre, During, and Post Spaceflight" with PI Dr. Eric Rivas -- and prior to that, Dr. Meghan Downs. See those projects for previous reporting. In this reporting period, we have coordinated with the Research Operations and Integration element to initiate the consenting of crewmembers for participation in this investigation. Two participants have signed informed consents and data collection is anticipated to start in 2023.
Bibliography Type:	Description: (Last Updated: 04/09/2026)
Abstracts for Journals and Proceedings	Downs ME, Ade C, Barstow TJ, Ryder J, Martin D, Abercromby A, Smith S, Norcross J, Goetchius E, Rivas E, Feiveson A. "Temporal changes in astronauts' muscle and cardiorespiratory physiology pre-, in-, and post-spaceflight. Poster." 2021 NASA Human Research Program Investigators' Workshop, Virtual, February 1-4, 2021. Abstracts. 2021 NASA Human Research Program Investigators' Workshop, Virtual, February 1-4, 2021. , Feb-2021
Abstracts for Journals and Proceedings	Rivas E, Ade C, Barstow T, Martin D, Abercromby A, Norcross J, Smith S, Young M. "Temporal changes in astronauts' muscle and cardiorespiratory physiology before, during, and after spaceflight. Poster." 2022 NASA Human Research Program Investigators' Workshop, Virtual, February 7-10, 2022. Abstracts. 2022 NASA Human Research Program Investigators' Workshop, Virtual, February 7-10, 2022. , Feb-2022
Abstracts for Journals and Proceedings	Strock N, Prejean B, Norcross J, Schlotman T, Frisco D, Young M, Abercromby A, Marshall-Goebel K. "Temporal changes in astronauts' muscle and cardiorespiratory physiology before, during, and after spaceflight. Poster." 2023 NASA Human Research Program Investigators' Workshop, "To the Moon: The Next Golden Age of Human Spaceflight", Galveston, TX, February 7-9, 2023. Abstracts. 2023 NASA Human Research Program Investigators' Workshop, "To the Moon: The Next Golden Age of Human Spaceflight", Galveston, TX, February 7-9, 2023. , Feb-2023
Articles in Peer-reviewed Journals	Ntoumani M, Dugué B, Rivas E, Gongaki K. "Thermoregulation and thermal sensation during whole-body water immersion at different water temperatures in healthy individuals: A scoping review." J Therm Biol. 2023 Feb;112:103430. Review. https://doi.org/10.1016/j.jtherbio.2022.103430 ; PMID: 36796887 , Feb-2023